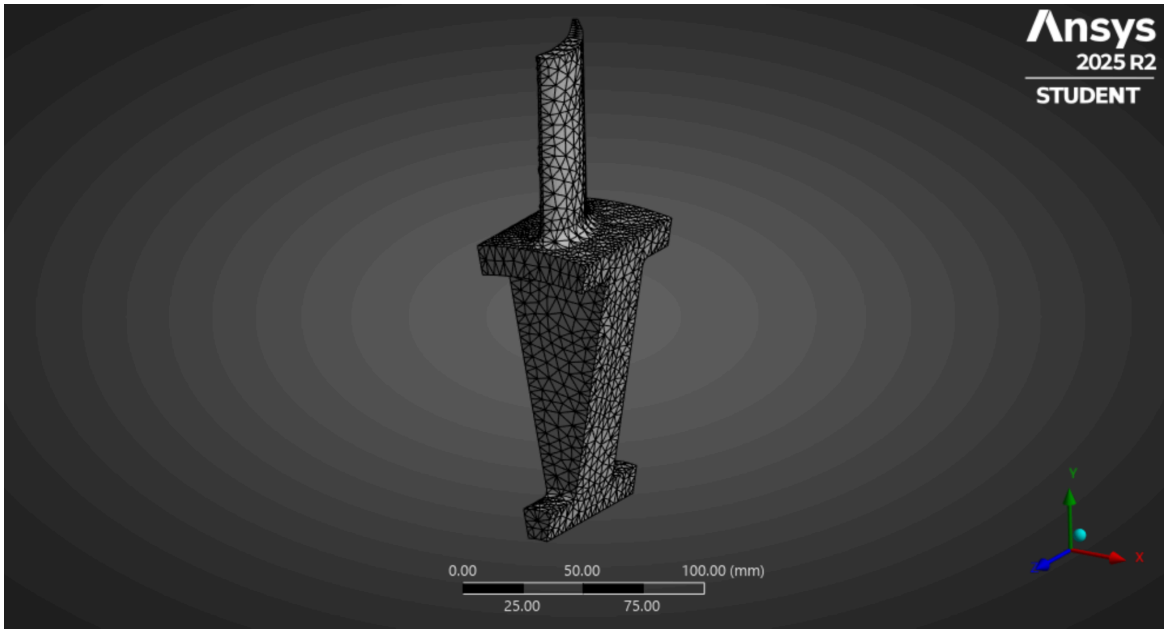
Matthew Heering

5/1/2026

Modal Analysis of a Turbine in Ansys

1.



Statistics	
Nodes	14998
Elements	8131
Show Detailed Statistics	No

Total DOF: $14998(Nodes)\ *\ 3(DOF/Node)\ =\ 44994DOF$

The essential boundary conditions determine the DOF of the nodes on the bore of the disk.

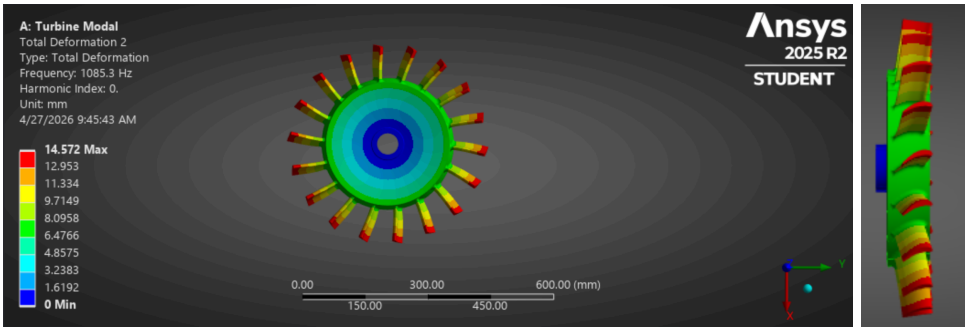
2.

a.

Tabular Data

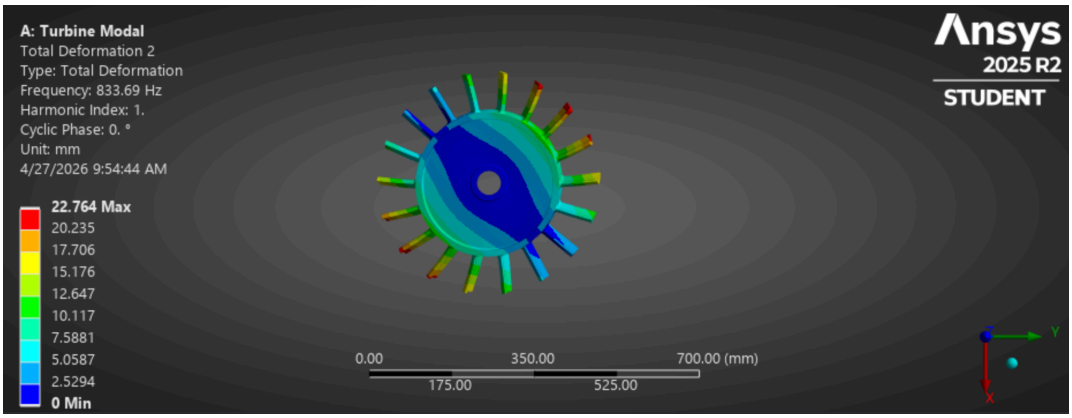
	Mode	Harmonic Index	Frequency [Hz]
1	1.	0.	957.34
2	2.	0.	1085.3
3	3.	0.	1676.2
4	1.	1.	833.69
5	2.	1.	833.69
6	3.	1.	1187.9
7	1.	2.	1082.5
8	2.	2.	1082.5
9	3.	2.	1282.4

b.



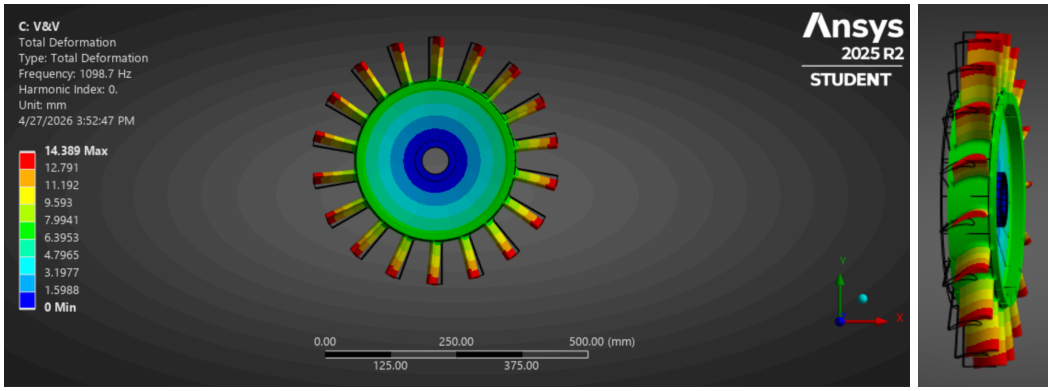
The absolute values of the displacement shown in the figure are not physically meaningful since modal analysis works by generating eigenvectors that are arbitrarily scaled. Thus, the meaningful result is not the absolute values of the displacement, but the shape and relative amplitudes.

c.



If you take the circumference and unravel it into a line, there will be one wave through the line for a harmonic index of 1. That's why you can see a single wave of displacement traveling through the disk in the photo and know that the harmonic index is not zero.

3.



With rotation (this case): $\omega_0 = 1098.7Hz$

Without rotation (2b): $\omega_0 = 1085.3Hz$

The natural frequency has increased with the effect of pre-stress included. This meets my expectation since the natural frequency for the same mode will increase since rotation stiffens the structure.